Frobenius Minimality, Minimum-Layer Uniqueness, and an Infinite Family of Huneke–Wiegand Counterexamples

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Provenance. Son Pham discovered the counterexample; Professor Craig Huneke independently verified it. This paper contributes certified Frobenius minimality, minimum-layer uniqueness, a separate explicit infinite family, and its uniform endomorphism, type, trace, conductor, stability, reduction, tangent-cone, fiber-cone, defining-ideal, and syzygy anatomy.

Abstract

Son Pham publicly identified a counterexample to the Huneke–Wiegand conjecture in the class of two-generated monomial ideals over symmetric numerical semigroup rings, subsequently verified by Professor Craig Huneke. We preserve that discovery priority and answer two different questions: how small can such an example be, and does the same semigroup/ideal class contain an explicit infinite family? We prove by proof-carrying exact computation that the least Frobenius number is 181, attained by Pham's semigroup at ideal shift 14. First, an implementation of the complete Blanco–Rosales tree exhausts all 48,954 symmetric semigroups with Frobenius number below 69 and all 1,503,391 associated gap cases. An independent Boolean route checks 1,156 fixed-pair UNSAT certificates for the same range. We then introduce a one-hot shift-selector CNF that asks one existence question per Frobenius number. Its DRAT certificates exclude every odd value from 69 through 179, while the formula at 181 is satisfiable and decodes exactly to the public semigroup and shift. A separately generated fixed-pair formula returns the same model. An independent audit rehashes all 228 files in the minimality search, totaling 760,081,739 bytes, rechecks both SAT models, and reproduces the strict-order manifest with no mismatch. Complete theorem-tree enumeration at $F = 69, 71, 73, 75$ provides an object-level cross-check on another 79,790 semigroups and 2,933,163 gap cases. A second projected enumeration proves that shift 14 is the only feasible shift at $F = 181$ and that the public membership vector is the only semigroup there. Both terminal DRAT proofs are freshly rechecked in an independent 12-file audit. Thus Pham's pair is the unique normalized pair attaining the Frobenius minimum in this class. Finally, for every integer $p \geq 4$ we construct a symmetric numerical semigroup Γ_p of multiplicity $24p$, Frobenius number $78p - 1$, conductor $78p$, and embedding dimension $11p$, for which $I_p = (t^{24p}, t^{30p})$ is nonprincipal and rigid. This family theorem is deductive: seven exact interval-sum identities prove closure, symmetry, and $D_s = E_s + E_s$; a finite sweep through $p = 300$ and an independently implemented audit are supporting checks. For every family member we also determine the value semigroup of the endomorphism ring exactly. It has multiplicity $24p$,

Frobenius number $54p-1$, conductor $54p$, genus $38p-1$, and embedding dimension $12p$. It is nonsymmetric, and the endomorphism ring is strictly larger than the original ring. Recent endomorphism-center criteria then show uniformly that the ideal remains rigid over its endomorphism ring but is not reflexive there, with the adjacent Ext and Tor obstruction groups nonzero. We further prove that this endomorphism semigroup has Cohen–Macaulay type and reduced type $10p$, hence maximal reduced type, and that its completed semigroup ring is not almost Gorenstein. Finally, the traces of the ideal and its endomorphism ring both equal the conductor of the finite birational extension. We determine that common value ideal exactly and prove the identity

$$\ell_{R_p}(R_p/(R_p:E_p)) = \ell_{R_p}(E_p/R_p) = p+1.$$

The equality of the two lengths is a general consequence of one-dimensional Gorenstein local duality, while the exact common ideal and its value $p+1$ are the family-specific data. We further prove that this conductor is nonstable and compute its complete one-step stability defect:

$$\ell_{R_p}(T_p^2/t^{4s}T_p) = 14p.$$

The full reduction sequence is also exact: $t^{4s}R_p$ is a minimal reduction of T_p with reduction number four, successive Sally-quotient lengths $23p-1, 14p, 2p, 1, 0$, and Hilbert–Samuel coefficients $e_0(T_p) = 24p$ and $e_1(T_p) = 39p$. The associated graded ring of T_p nevertheless has depth zero: its complete Valabrega–Valla module is concentrated in one degree, where it has length p. We compute the exact Hilbert series and show that its numerator has only positive coefficients despite this failure of Cohen–Macaulayness. We then determine the complete zeroth local cohomology: it is k^p, concentrated in degree zero and annihilated by the full homogeneous maximal ideal. Hence the tangent cone is Buchsbaum but not Cohen–Macaulay, with unbounded Buchsbaum invariant p; its quotient by the finite-length torsion is Cohen–Macaulay with an exact Hilbert series. Over the polynomial Noether normalization induced by the minimal reduction, we determine the entire graded module: the torsion consists of p exponent-one cyclic summands, while the free quotient has rank $24p$ and explicit shifts. This yields the complete minimal resolution, projective dimension one, regularity four, top-local-cohomology a-invariant three, and the exact parameter-section identity $\ell(G_p/x_pG_p) = 25p = e_0(T_p) + I(G_p)$. Finally, the exact identity $T_p^2 = \mathfrak{m}_pT_p$ identifies the special fiber canonically with $G_p/H^0(G_p)$. It is Cohen–Macaulay of type $10p+1$, but its Artinian socle lies in degrees two and four, so it is neither level nor Gorenstein. We then determine its defining ideal: in the polynomial ring on its $10p$ minimal degree-one generators, it has exactly $50p^2 - 17p$ minimal quadrics and the single additional cubic $X_0^2X_{3p} - X_p^3$, with no other minimal equation. Thus its relation type is three and it is not Koszul. The all-parameter component calculation is an exact Presburger verification, supported by a separately encoded graph audit; its solver trust boundary is stated explicitly. From the same exact h-vector, Artinian socle, and first Betti row, we then determine the homological edges over the full presentation ring. Its projective dimension is $10p-1$, its regularity is four,

$$\beta_{2,3} = \frac{2p(500p^2 - 330p + 31)}{3},$$

the complete last row has ranks $10p$ in shift $c+2$ and one in shift $c+4$ for $c = 10p-1$, and $\beta_{c-1,c+3} = 8p$. The canonical module has $10p$ minimal generators in degree -1 and one in degree -3. We then determine the first previously unknown interior strand. With the offset grading $\deg X_a = (1,a)$,

$$\beta_{2,4} = 8p, \qquad \beta_{3,4} = \frac{p(5p-1)(500p^2 - 440p + 47)}{2},$$

and the $8p$ second syzygies occur with multiplicity one exactly at offsets $\{3p+a : a \in \mathcal{E}_{p,1},\ a \geq 6p\}$. Relative squarefree-divisor chains and an integral unit matching give the upper bound; an exact quadratic colon and minimal mapping cone give the matching lower bound. Thus the formulas hold over every field. This determines one interior strand, not the remaining Betti table. The results remain confined to two-generated monomial ideals over symmetric numerical semigroup rings and do not classify arbitrary Gorenstein domains and modules.

1 Introduction

Let R be a one-dimensional Gorenstein local domain and let M be a nonzero finitely generated non-free module. The Huneke–Wiegand conjecture predicted nontrivial torsion in $M \otimes_R \operatorname{Hom}_R(M, R)$ [1]. For a numerical semigroup $\Gamma \subseteq \mathbb{N}$, the semigroup ring $\Bbbk[t^\Gamma]$ is Gorenstein exactly when Γ is symmetric. This specialization converts part of the module problem into finite additive combinatorics. García-Sánchez and Leamer proved broad positive classes and reported a NumericalSgps check for every symmetric semigroup with Frobenius number below 69 [2].

In 2026, Son Pham published a symmetric numerical semigroup and a two-generated monomial ideal for which the predicted torsion is absent [5]. Professor Craig Huneke supplied an independent verification archived with the public candidate record. That example has Frobenius number 181. The discovery settles the original conjecture negatively, but does not by itself say whether 181 is small, minimal, or typical.

This paper reports computer-assisted classification theorems and a deductive family theorem. Claims are labeled **[MV]** when they are verified by exact machine computation and independently checked artifacts, and **[D]** when derived in the text. No conjectural claim is needed for the main theorem.

Contributions.

1. We give a self-contained finite-window reduction and a direct proof that the selector CNF is equisatisfiable with existence of a rigid two-generated monomial ideal.
2. We reproduce the published $F < 69$ boundary by two independent complete routes: the Blanco–Rosales tree and checked DRAT proofs.
3. We certify UNSAT for every odd $F = 69, 71, \ldots, 179$, then recover the exact public model at $(F, s) = (181, 14)$.
4. We independently rehash and inspect the complete search artifact set, and cross-check the first four new odd Frobenius values by explicit theorem-tree enumeration.
5. We classify the complete minimum layer by projected enumeration: the only feasible shift at $F = 181$ is 14, and the public semigroup is the only membership vector at that shift.
6. We identify two structural boundaries: the public semigroup lies outside the generalized arithmetic-sequence positive family, and the visible fixed-offset generator blocks do not extend to a naive parametric family.
7. We use the failed fixed-width ansatz to derive and prove an explicit infinite family for every integer $p \geq 4$, with a separate exact implementation and independent audit.
8. We compute the endomorphism overring of every family member and prove that the nonreflexive Ext/Tor escape identified in the public seed persists uniformly.
9. We compute the complete pseudo-Frobenius set of every endomorphism semigroup, proving unbounded maximal reduced type and a uniform non-almost-Gorenstein boundary.

10. We compute both trace ideals and the conductor exactly, obtaining a common ideal whose colength equals the birational extension length, namely $p+1$.
11. We separate that general duality balance from the family-specific calculation, prove that the common conductor is nonstable, and compute its exact stability defect $14p$.
12. We determine every later power of the conductor, proving reduction number four, the full Sally-quotient profile, and Hilbert–Samuel coefficients $(e_0, e_1) = (24p, 39p)$.
13. We prove that the conductor tangent cone has depth zero, determine its unique Valabrega–Valla defect of length p, and compute its coefficientwise-positive Hilbert numerator exactly.
14. We identify that defect with the complete zeroth local cohomology, prove that the full homogeneous maximal ideal annihilates it, and obtain a Buchsbaum non-Cohen–Macaulay family with unbounded Buchsbaum invariant p and an exact Cohen–Macaulay quotient series.
15. We determine the complete graded module over the minimal-reduction Noether normalization, its minimal resolution and Betti numbers, regularity four, a-invariant three, and the exact parameter-section identity $\ell(G_p/x_pG_p) = 25p = e_0(T_p) + I(G_p)$.
16. We prove $T_p^2 = \mathfrak{m}_pT_p$, identify the conductor special fiber naturally with $G_p/H^0(G_p)$, and compute its type $10p+1$ and nonlevel Artinian socle.
17. We determine the complete defining ideal of that special fiber: $50p^2 - 17p$ minimal quadrics and one primitive cubic, giving relation type three and a non-Koszul family.
18. We determine exact homological edges over its full presentation ring: projective dimension and regularity, the linear first-syzygy count, the complete last row, one penultimate extremal entry, and the canonical-module generator degrees.
19. We determine the first interior Betti strand over every field: $\beta_{2,4} = 8p$ with complete multiplicity-free offset support, and the resulting exact formula for $\beta_{3,4}$.

2 The finite semigroup problem

Definition 2.1. A *numerical semigroup* is a cofinite additive submonoid $\Gamma \subseteq \mathbb{N}$. Its Frobenius number is $F(\Gamma) = \max(\mathbb{N} \setminus \Gamma)$. It is *symmetric* when

$$n \in \Gamma \quad \Longleftrightarrow \quad F(\Gamma) - n \notin \Gamma \qquad (0 \le n \le F(\Gamma)).$$

Fix a symmetric semigroup Γ with Frobenius number F and a positive gap $s \notin \Gamma$. Normalize the corresponding two-generated relative ideal to $(0, s)$. Define

$$E_s = \{n \in \mathbb{N} : n, n+s \in \Gamma\},$$
$$D_s = \{n \in \mathbb{N} : n, n+s, n+2s \in \Gamma\}.$$

The García-Sánchez–Leamer reduction identifies the counterexample condition with

$$D_s = E_s + E_s. \tag{1}$$

Equivalently, every length-two arithmetic sequence of step s in Γ decomposes as a sum of two length-one sequences [2, 4].

Lemma 2.2 (Automatic inclusion, [D])**.** *For every numerical semigroup* Γ *and every positive integer* s*,* $E_s + E_s \subseteq D_s$*.*

Proof. If $a, b \in E_s$, then $a, a+s, b, b+s \in \Gamma$. Closure gives $a+b \in \Gamma$, $a+b+s = a+(b+s) \in \Gamma$, and $a+b+2s = (a+s)+(b+s) \in \Gamma$. Hence $a+b \in D_s$. □

Lemma 2.3 (Finite window, [D])**.** *Equation* (1) *holds if and only if* $D_s \cap [0, 2F+1] \subseteq E_s + E_s$*.*

Proof. Only the reverse inclusion needs checking by Lemma 2.2. Every integer at least $F+1$ belongs to Γ, hence to E_s and D_s. Let $m = \min E_s$; then $m \le F+1$. For every $n \ge m+F+1$, both m and $n-m$ lie in E_s, so $n \in E_s + E_s$. Since $m+F+1 \le 2F+2$, every $n \ge 2F+2$ is automatic. The displayed finite window is therefore complete. □

Lemma 2.4 (Parity, [D])**.** *The Frobenius number of a symmetric numerical semigroup is odd.*

Proof. If F were even, symmetry at $n = F/2$ would require $F/2$ to belong to Γ exactly when it does not belong to Γ. □

3 The public model and independent calibration

Pham's public semigroup is

$$\Gamma = \langle 56, 57, 58, 63, 64, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, \\ 87, 89, 90, 93, 95, 96, 97\rangle,$$

with the ideal generated by t^{56} and t^{70}, hence normalized shift $s = 14$ [5]. An independent Singular/4ti2 calculation and a separate standard-library finite route verify $F = 181$, conductor 182, genus 91, symmetry, and equality (1) **[MV]**. The Singular route uses a 322-generator toric standard basis; both reduced ideal differences vanish. The hypersurface control $\langle 4, 5\rangle$ rejects equality with explicit residues **[MV]**.

The endomorphism overring of the ideal has value semigroup

$$v(\mathrm{End}_R(I)) = \Gamma \cup \{101, 107, 181\},$$

with Frobenius number 125, genus 88, and type 24 **[MV]**. This explains why positive criteria that require a Gorenstein endomorphism ring do not apply, but it does not imply minimality.

4 Complete reproduction below 69

For fixed odd F, irreducible numerical semigroups are exactly the symmetric ones. Blanco and Rosales construct all of them as a rooted tree [3]. Starting from

$$C(F) = \{0, (F+1)/2, (F+1)/2+1, \ldots\} \setminus \{F\},$$

a child replaces a suitable minimal generator $x > F/2$ by $F - x$. Their theorem supplies exact child conditions, completeness, and a unique parent.

Our direct implementation validates every node for closure, symmetry, genus, minimal generators, and its theorem parent. It then checks every gap by Lemma 2.3 and retains a first element of $D_s \setminus (E_s + E_s)$ whenever equality fails. The full range $F = 1, 3, \ldots, 67$ contains 48,954 semigroups and 1,503,391 gap cases; none is rigid [**MV**]. At $F = 11$ the implementation returns the six nodes listed by Blanco–Rosales, while an invalid mutation is rejected by closure at $(1, 1)$.

An independent route generates one fixed-pair DIMACS formula for each (F, s), including shifts that are forced not to be gaps. CaDiCaL emits a DRAT proof for every UNSAT result, and DRAT-trim checks each proof against the exact CNF [20, 21]. All 1,156 formulas are UNSAT and independently verified [**MV**]. A committed auditor rehashes 4,624 external files totaling 211,671,241 bytes with zero mismatch.

5 One selector formula per Frobenius value

The fixed-pair route duplicates the same semigroup constraints once per shift. We instead introduce Boolean membership variables h_n for $0 \le n \le F$ and one-hot variables q_s for $1 \le s \le F$.

5.1 Semigroup and selector clauses

The CNF contains units h_0 and $\neg h_F$, the symmetry clauses

$$(h_n \vee h_{F-n}) \wedge (\neg h_n \vee \neg h_{F-n}),$$

and closure clauses $\neg h_a \vee \neg h_b \vee h_{a+b}$ for $a + b \le F$. The selectors satisfy one at-least-one clause, all pairwise at-most-one clauses, and $q_s \Rightarrow \neg h_s$. Thus every model chooses exactly one gap.

For $0 \le n \le 2F + 1$, guarded Tseitin clauses impose, under the selected q_s,

$$\begin{aligned} e_n &\longleftrightarrow h_n \wedge h_{n+s}, \\ d_n &\longleftrightarrow h_n \wedge h_{n+s} \wedge h_{n+2s}, \end{aligned}$$

where h_j is the constant true for $j > F$. For each unordered decomposition $n = a + b$, a variable $p_{n,a,b}$ is equivalent to $e_a \wedge e_b$. Finally,

$$d_n \Rightarrow \bigvee_{a+b=n} p_{n,a,b}.$$

The automatic inclusion from Lemma 2.2 need not be duplicated in the formula.

Proposition 5.1 (Selector equisatisfiability, [D])**.** *For positive odd F, the selector CNF is satisfiable if and only if there exist a symmetric numerical semigroup Γ with Frobenius number F and a gap s satisfying $D_s = E_s + E_s$.*

Proof. The semigroup clauses decode exactly a symmetric semigroup of Frobenius number F, and the one-hot clauses decode a unique gap. Because exactly one guard is true, all e_n and d_n equal the definitions for that gap. The p equivalences and final disjunctions impose $D_s \subseteq E_s + E_s$ through $2F + 1$. Lemmas 2.2 and 2.3 give global equality. Conversely, from a rigid pair assign h and q from the pair, assign e, d by their definitions, and assign each p by its conjunction. Every clause is then true. □

The selector encoding passes two adversarial calibrations [**MV**]. Every odd $F \le 67$ is UNSAT with an accepted proof, while $F = 181$ is SAT. The latter formula has 34,397 variables and 363,912 clauses, selects $s = 14$, and decodes exactly to the public membership vector. A newly generated fixed-pair formula at $(181, 14)$ returns the same vector.

6 The certified search and minimality theorem

We scan odd F in strict increasing order. Table 1 summarizes the closed run.

Table 1: Selector-CNF minimality search.

Quantity	Value
Odd Frobenius values requested	57 (69 through 181)
UNSAT with accepted DRAT proof	56 (69 through 179)
Validated SAT models	1 (181, shift 14)
UNKNOWN or timeout	0
Wall time	4,303.86 seconds
CNF bytes	169,745,433
DRAT proof bytes	586,785,257
Slowest solve	$F = 175$, 218.83 seconds

Theorem 6.1 (Frobenius minimality, [MV+D]). *Among symmetric numerical semigroups admitting a nonprincipal two-generated monomial ideal that is a counterexample to the Huneke–Wiegand prediction, the least Frobenius number is* 181. *It is attained by Pham's public example at shift* 14.

Proof. By Lemma 2.4, only odd Frobenius values occur. Complete tree enumeration and the independent fixed-pair proof suite exclude every odd $F \leq 67$. Proposition 5.1 and the accepted selector DRAT proofs exclude every odd F from 69 through 179. The validated selector and fixed-pair models at $(181, 14)$ provide existence. These cases exhaust all smaller positive Frobenius values. □

The search audit independently rehashes all 228 expected CNF, proof, solver-log, and checker-log files, totaling 760,081,739 bytes. It also reconstructs both SAT membership masks, validates closure, symmetry, and rigidity, checks strict odd ordering, and recomputes aggregate SHA-256

`0f580de2707a00fdd52e1b3c04e7767b97ce7b0a826593b119e9a49ae04da743`.

No file is missing, no hash differs, and no status marker or semantic check fails [**MV**].

7 Complete classification at the minimum

The selector search proves existence at $F = 181$ but, by itself, does not classify every model at that value. We close this gap in two projected layers. First, after discovering a valid selector model at shift s, we add the unit clause $\neg q_s$. Second, for each feasible shift we enumerate the fixed-shift formula and block each complete membership projection.

Lemma 7.1 (Projected blocker, [D]). *Let $x_1, \ldots, x_k$ be distinct Boolean variables with complete assignment a. The clause*

$$B(a) = \bigvee_{a_i=1} \neg x_i \ \vee \bigvee_{a_i=0} x_i$$

is false exactly on assignment a. Its truth is independent of every variable outside the projection.

Proof. Every displayed literal is false under a. Any other complete assignment differs at some coordinate j, where the corresponding displayed literal is true. Variables absent from the clause cannot affect either statement. □

For semigroup enumeration, $B(a)$ contains one signed literal for every $h_0, \dots, h_{181}$. Consequently, it excludes one mathematical membership vector and every auxiliary extension of that vector, while preserving every different semigroup. Blocking only the positive coordinates would not have this property.

Theorem 7.2 (Minimum-layer uniqueness, [MV+D])**.** *Up to translation and interchange of the two monomial generators, Pham's pair is the unique nonprincipal two-generated monomial-ideal counterexample over a symmetric numerical semigroup ring with Frobenius number* 181*.*

Proof. The unblocked selector formula returns the exact public pair at shift 14. After adding $\neg q_{14}$, the formula is UNSAT and its DRAT proof is accepted, so no other shift is feasible. The fixed $(181, 14)$ formula returns the exact public membership vector. After adding its complete 182-literal blocker from Lemma 7.1, that formula is UNSAT and its DRAT proof is accepted. Thus no second semigroup exists at the only feasible shift. Translation and interchange reduce any two monomial generators to one positive normalized shift, which proves the statement. □

Table 2: Certificate summary for the complete $F = 181$ layer.

Quantity	Value
Feasible shifts	1, namely $\{14\}$
Normalized rigid pairs	1
Distinct membership vectors	1
Support terminal proof bytes	45,867,741
Fixed-class terminal proof bytes	1,608,691
Audited external files	12
Audited external bytes	63,609,504
Fresh proof-recheck time	232.88 seconds

The independent auditor reconstructs all four CNFs byte-for-byte, decodes both SAT logs, reruns closure, symmetry, minimal-generator and full-window/tail rigidity checks, and freshly checks both terminal proofs [**MV**]. It accounts for every external file and reproduces manifest SHA-256

`02d1265b75e886bc6ec693e60b367b661476170c755e38a220342396564ef5d3`.

The compact classification aggregate is

`b37d9410fa07c00f22ccc0f83796644db3b44f0dd654c06933aa5d22cb1ed788`.

8 Independent object-level checks beyond the old frontier

The theorem-tree route was continued beyond 67 without consulting SAT models. Table 3 lists the first four new odd values.

This is not needed to trust a DRAT derivation, but it attacks the custom encoding at the object level and supplies explicit failed-rigidity witnesses for every tested gap. The routes agree.

Table 3: Complete Blanco–Rosales tree cross-checks beyond the published boundary.

F	Symmetric semigroups	Gap cases	Rigid cases	Seconds
69	11,276	394,660	0	137.00
71	19,812	713,232	0	235.37
73	25,405	939,985	0	333.21
75	23,297	885,286	0	330.10
Total	79,790	2,933,163	0	1,035.68

9 From a failed ansatz to an infinite family

Landeros et al. prove the positive result for semigroups generated by a generalized arithmetic sequence $\langle a, ah+d, \ldots, ah+kd\rangle$ [4]. The public semigroup cannot lie in this class **[D]**. Its multiplicity forces $a = 56$. Since $57 \in \Gamma$ and no sum of two positive semigroup elements is below 112, the presentation forces $h = d = 1$. Since $63 \in \Gamma$, it would then include $59, 60, 61, 62$, but all four are gaps.

The seed also displays $56 = 4s$ and $181 = 13s - 1$ at $s = 14$, with generator blocks

$$4s + \{0, 1, 2, 7, 8\}, \qquad 5s + [0, s-1], \qquad 6s + \{3, 5, 6, 9, 11, 12, 13\}.$$

A predeclared exact sweep refutes the naive fixed-offset lift **[MV]**. Only $s = 14$ passes among even $s \le 100$. At $s = 16$, the generated semigroup has $F = 205$ rather than 207; at $s = 28$ the predicted Frobenius value returns but symmetry fails. This negative result prevents the minimality theorem from being overinterpreted as evidence of an immediate family.

We next retained only the structural scaffold $m = 4s$, $F = 13s-1$, the full interval $[5s, 6s-1] \subseteq \Gamma$, and rigidity at shift s. A proof-carrying Boolean search certifies that the scaffold is infeasible at $s = 16, 18$ and returns independently validated models for every even $s = 20, 22, \ldots, 40$ **[MV]**. The eleven non-seed models justified a separate extraction stage, but did not themselves prove recurrence. The first fixed-width interval formula passed three parameters and then failed: at $s = 38$, $9s + 7$ lies in D_s but not in $E_s + E_s$. For every later parameter in that formula, carried sums from the first block end at residue 6, while the next summand block begins at residue at least 8. This uniform obstruction suggests widening the first interval with the parameter.

9.1 The parametric blocks

Fix an integer $p \ge 4$ and put $s = 6p$. All intervals below are integer intervals. Define subsets of $[0, s-1]$ by

$$\begin{aligned}
A_p &= [0, p] \cup [3p, 4p-2], \\
B_p &= ([p+1, 3p-1] \setminus \{2p-1\}) \cup \{4p\} \cup [5p-1, 6p-1], \\
C_p &= [0, 2p] \cup [3p, 5p-2].
\end{aligned}$$

Define $\Gamma_p \subseteq \mathbb{N}$ by the membership blocks

$$\{0\}, \quad 4s + A_p, \quad [5s, 6s-1], \quad 6s + B_p, \quad 8s + C_p, \quad [9s, 13s-2], \quad [13s, \infty), \tag{2}$$

with every unlisted nonnegative integer below $13s$ a gap.

For subsets of $[0, s-1]$, let $\mathrm{low}(X+Y)$ denote the sums below s, and let $\mathrm{carry}(X+Y)$ denote the sums at least s after subtraction of s.

Lemma 9.1 (Residue identities, [D]). *For every $p \geq 4$,*

$$\begin{aligned} \mathrm{low}(A_p + A_p) &= C_p, & \mathrm{carry}(A_p + A_p) &= [0, 2p-4], \\ \mathrm{low}(A_p + B_p) &= [p+1, 6p-1], & \mathrm{carry}(A_p + B_p) &= [0, 4p-3], \\ \mathrm{low}(B_p + B_p) &= [2p+2, 6p-2], & \mathrm{carry}(B_p + B_p) &= [0, 6p-2], \\ \mathrm{low}(A_p + C_p) &= [0, 6p-2]. \end{aligned}$$

Moreover,

$$[0, 2p-4] \cup [p+1, 6p-1] = [0, 6p-1], \tag{3}$$

$$[0, 4p-3] \cup [2p+2, 6p-2] = [0, 6p-2]. \tag{4}$$

Proof. Split A_p into its two displayed intervals and split the punctured component of B_p at $2p-1$. Pairwise interval addition, followed by separating sums below and above $s = 6p$, gives the seven displayed identities. The singleton $4p$ fills the endpoint removed from the middle of B_p. All interval joins are immediate except (3): its two pieces meet exactly when $2p-3 \geq p+1$, equivalently $p \geq 4$. For (4), the overlap condition is $4p-2 \geq 2p+2$, which also holds for $p \geq 4$. ☐

Theorem 9.2 (Infinite family, [D]). *For every integer $p \geq 4$, the set Γ_p in (2) is a symmetric numerical semigroup with multiplicity $24p$, Frobenius number $78p-1$, conductor $78p$, and embedding dimension $11p$. Let*

$$R_p = \Bbbk[t^{\Gamma_p}]_{(t^n : n \in \Gamma_p \setminus \{0\})}, \qquad I_p = (t^{24p}, t^{30p})R_p.$$

Then R_p is Gorenstein and I_p is nonprincipal and rigid. Hence the pairs (R_p, I_p) form an infinite family of counterexamples in the two-generated monomial-ideal class.

Proof. We first prove the semigroup assertions. The only sums of positive displayed members that can stay below the conductor have starting levels among $4, 5, 6, 8$. Lemma 9.1 handles the non-full target blocks: $A_p + A_p$ produces $8s + C_p$ and part of level 9; $A_p + B_p$, $B_p + B_p$, and $A_p + C_p$ land exactly inside the declared high blocks and never produce the Frobenius residue $s-1$ in level 12. Sums involving the full level 5 block either fill a declared level or enter the conductor tail. Thus Γ_p is closed.

Conversely, the lower three blocks generate every displayed member. The identity $\mathrm{low}(A_p + A_p) = C_p$ generates level 8. The presence of $0 \in A_p$ makes $(4s + A_p) + [5s, 6s-1]$ fill level 9, and two level-5 elements fill level 10 and residues $0, \ldots, s-2$ of level 11; the residue $s-1 \in B_p$ fills the remaining point. Three elements from $4s + A_p$ generate residues $0, \ldots, 3p$ of level 12, while $\mathrm{low}(B_p + B_p) = [2p+2, s-2]$ completes that level through $13s-2$. Adding $4s$ to $[9s, 13s-2]$ gives $[13s, 17s-2]$, and $(4s+1) + (13s-2) = 17s-1$. These are $4s$ consecutive generated integers, so repeated addition of the multiplicity gives the entire tail.

Therefore the multiplicity is $4s = 24p$, the conductor is $13s = 78p$, and the Frobenius number is $13s - 1 = 78p - 1$. Every member of the three lower blocks lies below $7s$, whereas a sum of two positive members is at least $8s$; all are minimal generators. Since $|A_p| = 2p$, $|B_p| = 3p$, and the full block has size $6p$, the embedding dimension is $11p$.

Symmetry is blockwise about $F = 13s - 1$. The residue-reflection complement of A_p is exactly C_p. The full level-5 block reflects to the empty level-7 block. In B_p, exactly one residue from each pair $\{r, s-1-r\}$ occurs: the puncture $2p-1$ pairs with the singleton $4p$, and the remaining displayed intervals pair with their omitted intervals. The gaps below $4s$ reflect to the full range $[9s, 13s-2]$, and 0 reflects to the gap F. Hence Γ_p is symmetric, so R_p is Gorenstein.

It remains to prove rigidity. Normalize I_p to $J_p = (1, t^s)$ and use E_s, D_s from Section 2. Direct inspection of (2) gives the nontrivial residue layers

$$\begin{aligned} E_4 &= A_p, & E_5 &= B_p, & E_8 &= C_p, \\ E_9 = E_{10} &= [0, s-1], & E_{11} = E_{12} &= [0, s-2], \\ D_8 &= C_p, & D_9 &= [0, s-1], \\ D_{10} = D_{11} = D_{12} &= [0, s-2]. \end{aligned}$$

All lower D layers are empty; in particular D_4 is empty because $A_p \cap B_p = \varnothing$. Lemma 9.1 and (3)–(4) compute the five load-bearing layers of $E_s + E_s$:

$$D_8 = \mathrm{low}(A_p + A_p), \quad D_9 = \mathrm{carry}(A_p + A_p) \cup \mathrm{low}(A_p + B_p),$$

$$D_{10} = \mathrm{carry}(A_p + B_p) \cup \mathrm{low}(B_p + B_p), \quad D_{11} = \mathrm{carry}(B_p + B_p), \quad D_{12} = \mathrm{low}(A_p + C_p).$$

At levels 13 and 14, the zero residue in A_p translates full high E blocks. At levels 15 and 16, those high blocks omit only $s-1$, and the residues $0, 1 \in A_p$ fill both the interval and its final endpoint. Level 17 is filled by $E_4 + E_{13}$; levels 18, 19, and 20 by $E_9 + E_9$, $E_9 + E_{10}$, and $E_{10} + E_{10}$; and level 21 by $E_8 + E_{13}$ because $0 \in C_p$. Every level from 22 onward is filled by $E_9 + E_q$ with $q \geq 13$, where the conductor makes E_q full. Thus $D_s \subseteq E_s + E_s$; Lemma 2.2 gives equality.

Finally, s is a gap. If J_p were principal, write $J_p = xR_p$. Since $1 \in J_p$, there is $r \in R_p$ with $xr = 1$, so $rJ_p = R_p$ and $(r, rt^s) = R_p$. In the local ring, one of these two generators must be a unit. Positive valuation excludes rt^s, so r is a unit and $t^s \in R_p$, a contradiction. The ideal is therefore nonprincipal and rigid. The multiplicities $24p$ are distinct, so the family contains infinitely many distinct rings. □

9.2 Exact support and boundaries

The proof of Theorem 9.2 is independent of a finite sweep. As adversarial support, the implementation evaluated every $p = 2, 3, \ldots, 300$ [**MV**]. Exactly the predicted boundary values $p = 2, 3$ fail, while all 297 values $p \geq 4$ pass the seven identities, closure, generation, symmetry, invariants, and full-window/tail rigidity. The instances $p = 4, 5$ reproduce independently generated Boolean models byte-for-byte. Two corrupted positive controls are rejected. A separate auditor reconstructs all 297 positive membership hashes from the formula and performs fresh semantic checks at $p = 4, 5, 17, 73, 151, 300$. The campaign aggregate is

`81d5a8eb6cf2e848807323e3b0bdba58c464779d25cdc788cef027585540dce2`.

The independent-audit aggregate is

`eb2aaf17650ed99f4e220a43c53bdd8835c82688a37567bb154c30a1ae520ce9`.

The family remains outside the generalized-arithmetic-sequence positive class [**D**]. Membership of $4s + 1$ forces unit step, while membership of $4s + 3p$ would then force the actual gap $4s + p + 1$. Theorem 9.2 does not say that every model in the Boolean scaffold belongs to this family, or that $p = 4$ is a global threshold outside the displayed formula.

10 Uniform endomorphism overrings

Normalize I_p to $J_p = (1, t^s)$ and write

$$V_p = v(J_p) = \Gamma_p \cup (s + \Gamma_p), \qquad \Lambda_p = v(\mathrm{End}_{R_p}(J_p)).$$

Define

$$Q_p = [p+1, 2p-2] \cup \{2p, 4p\}.$$

Theorem 10.1 (Uniform endomorphism anatomy, [D])**.** *For every integer $p \geq 4$,*

$$\Lambda_p = \Gamma_p \cup (7s + Q_p) \cup \{13s - 1\}.$$

This numerical semigroup has multiplicity $24p$, Frobenius number $54p - 1$, conductor $54p$, genus $38p - 1$, and embedding dimension $12p$. It is nonsymmetric, so the localized semigroup ring $E_p = \mathrm{End}_{R_p}(J_p)$ is not Gorenstein. Moreover,

$$\begin{aligned} &\mathrm{Ext}^1_{E_p}(J_p, J_p) = 0, && J_p \text{ is not reflexive over } E_p, \\ &\mathrm{Ext}^1_{E_p}(J_p, E_p) \neq 0, && \mathrm{Ext}^2_{E_p}(J_p, J_p) \neq 0, \\ &\mathrm{Tor}^{R_p}_1(J_p, E_p) \neq 0. && \end{aligned}$$

Proof. Let G_k and V_k denote the residue sets at levels ks in Γ_p and V_p, respectively. From (2),

$$V_0 = V_1 = \{0\}, \quad V_2 = V_3 = \varnothing, \quad V_4 = A_p, \quad V_5 = V_6 = [0, s-1], \quad V_7 = B_p, \quad V_8 = C_p,$$

and $V_k = [0, s-1]$ for every $k \geq 9$. An exponent $n = ks + r$ stabilizes J_p exactly when both n and $n + s$ belong to V_p. Thus the level-k residue set of Λ_p is $V_k \cap V_{k+1}$.

All adjacent intersections reproduce the corresponding Γ_p block except at level 7 and the old Frobenius point. Directly,

$$B_p \cap C_p = \big([p+1, 2p] \setminus \{2p-1\}\big) \cup \{4p\} = Q_p.$$

Every level from 9 onward is full, so the only other added value is $13s - 1$. This proves the formula for Λ_p.

The first positive value remains $4s = 24p$. Level 8 still omits residue $s - 1$, while level 9 and all later levels are full. Hence the Frobenius number is $9s - 1 = 54p - 1$ and the conductor is $9s = 54p$. Counting gaps by levels gives

$$(4s - 1) + (s - |A_p|) + (s - |B_p|) + (s - |Q_p|) + (s - |C_p|) = 38p - 1.$$

A symmetric semigroup with this Frobenius number would have genus $27p$, so Λ_p is nonsymmetric.

The $11p$ generators of Γ_p and the p new values $7s + Q_p$ all lie below twice the multiplicity, hence are minimal. For any $q \in Q_p$, the level-5 block contains $5s + (s - 1 - q)$ and

$$(7s + q) + (5s + (s - 1 - q)) = 13s - 1.$$

Thus the old Frobenius point is generated and there are exactly $12p$ minimal generators.

Finally, R_p is a one-dimensional Gorenstein local domain and J_p is faithful, torsion-free, two-generated, nonprincipal, and rigid by Theorem 9.2. The ring E_p is commutative and is strictly larger than R_p because Q_p is nonempty. Proposition 4.1(2) of Dey and Lyle makes rigidity descend to E_p; the contrapositives of their Theorems 4.2–4.4 give the displayed nonreflexivity, Ext, and Tor conclusions [6]. □

The symbolic argument is independent of a parameter sweep. As adversarial support, two exact implementations agree for every $p = 4, \ldots, 300$ [**MV**]. A separate auditor rehashes all 297 rows, reconstructs complete semantic windows at $p = 4, 5, 17, 73, 151, 300$, and rejects a missing Q_p value and a missing terminal singleton. The deterministic campaign and audit aggregates are

`e21926a689178a6c70b3b6e8319053edd0fd13f164ced9565d3b976e6159c0b0`

and

`2ed711045ad83a3b47fb3e71d4c75ae9bfa9be1a5dd9a8c4072d5f170510343b`.

11 Pseudo-Frobenius anatomy and reduced type

For a numerical semigroup H, write

$$\mathrm{PF}(H) = \{f \notin H : f + h \in H \text{ for every } h \in H \setminus \{0\}\}.$$

Its Cohen–Macaulay type is $|\,\mathrm{PF}(H)|$. For complements inside $[0, s-1]$, define

$$\begin{aligned} B_p^c &= [0,p] \cup \{2p-1\} \cup [3p, 4p-1] \cup [4p+1, 5p-2], \\ Q_p^c &= [0,p] \cup \{2p-1\} \cup [2p+1, 4p-1] \cup [4p+1, 6p-1], \\ C_p^c &= [2p+1, 3p-1] \cup [5p-1, 6p-1]. \end{aligned}$$

Theorem 11.1 (Type anatomy, [D]). *For every integer $p \geq 4$,*

$$\mathrm{PF}(\Lambda_p) = (6s + B_p^c) \cup (7s + Q_p^c) \cup (8s + C_p^c).$$

Consequently the type and reduced type of $\Bbbk[[\Lambda_p]]$ are both $10p$. Thus the ring has maximal reduced type. The semigroup is not almost symmetric, and $\Bbbk[[\Lambda_p]]$ is not almost Gorenstein.

Proof. The multiplicity of Λ_p is $4s$ and its conductor is $9s$. Every gap in the final multiplicity window $[5s, 9s-1]$ is therefore pseudo-Frobenius. Since level 5 is full, these gaps are precisely the complements of B_p, Q_p, C_p at levels 6, 7, and 8. Their cardinalities are $3p, 5p, 2p$.

It remains to exclude a pseudo-Frobenius number below $5s$. The full level-5 generator block gives explicit witnesses. For a gap r at level 0 with $r > 0$, add $5s + (s - r)$ to reach the level-6 gap of residue zero. At level 1, use $5s$ when $r = 0$ and $5s + (s - r)$ otherwise. At levels 2 and 3, add $5s + (s - 1 - r)$ to reach residue $s - 1$ in a missing later block.

For a level-4 gap $4s + r$, if $r \notin C_p$ add the generator $4s$. The remaining residues lie in

$$C_p \setminus A_p = [p+1, 2p] \cup [4p-1, 5p-2].$$

In the first interval add $4s + (2p + 1 - r)$; in the second add $4s + (5p - 1 - r)$. Both added residues lie in $[1, p] \subset A_p$, while the resulting level-8 residue is a gap of C_p. Thus no lower gap is pseudo-Frobenius, proving the formula.

Reduced type counts the gaps in the last multiplicity window, so it is also $10p$ [7]. Finally, using $g(\Lambda_p) = 38p - 1$ and $F(\Lambda_p) = 54p - 1$,

$$2g - (F + \text{type}) = 12p - 1 > 0.$$

The almost-symmetric genus identity therefore fails, giving the final claims. □

Two exact implementations confirm the displayed set for all $p = 4, \dots, 300$ **[MV]**. One checks every gap against the full minimal-generator set; the other extracts maximal elements of the Apéry poset modulo $4s$. The campaign and independent-audit aggregates are

`9bed38fb1c786c3740e000dde7ea7d79a7e7c83fa584ff12fc2c4623b5d503ec`

and

`0315c4c22c41e0d2b8a5abb27f717a4d4a6f7356ef30ca388206d739e0de2c37`.

12 Exact trace and conductor ideals

Let $E_p = \mathrm{End}_{R_p}(J_p) = \Bbbk[[\Lambda_p]]$ and define the reflected block

$$H_p = (s-1) - Q_p = \{2p-1, 4p-1\} \cup [4p+1, 5p-2].$$

Theorem 12.1 (Exact trace and conductor, [D])**.** *For every integer $p \geq 4$,*

$$\mathrm{tr}_{R_p}(J_p) = R_p : E_p = \mathrm{tr}_{R_p}(E_p) = T_p,$$

where

$$\begin{aligned} v(T_p) = {} & (4s + A_p) \cup (5s + (A_p \cup B_p)) \cup (6s + B_p) \cup (8s + C_p) \\ & \cup [9s, 13s-2] \cup [13s, \infty). \end{aligned}$$

Moreover,

$$\ell_{R_p}(R_p/T_p) = \ell_{R_p}(E_p/R_p) = p + 1.$$

Proof. Because $1 \in J_p$,

$$v(R_p : J_p) = W_p = \{n \in \Gamma_p : n + s \in \Gamma_p\}.$$

Intersecting adjacent blocks of (2) gives

$$(W_4, W_5, W_6, W_7, W_8) = (A_p, B_p, \varnothing, \varnothing, C_p),$$

with full blocks at levels 9 and 10, blocks $[0, s-2]$ at levels 11 and 12, and full blocks from level 13 onward. Since W_p is a Γ_p-ideal,

$$v(\mathrm{tr}_{R_p}(J_p)) = W_p \cup (s + W_p),$$

which is exactly the displayed set $v(T_p)$.

For the conductor $R_p : E_p$, only the values in

$$\Lambda_p \setminus \Gamma_p = (7s + Q_p) \cup \{13s - 1\}$$

add constraints. The terminal singleton excludes $n = 0$ and imposes no condition on positive $n \in \Gamma_p$. Every value at level 4 passes the $7s + Q_p$ translates: a failure in the sole level-12 gap would require $a + q = 2s - 1$, but $a + q \leq 8p - 2 < 12p - 1$. At level 5, residue r fails exactly when $r + q = s - 1$ for some $q \in Q_p$, hence exactly when $r \in H_p$. All values from level 6 onward translate into the conductor tail. Since

$$[0, s-1] \setminus H_p = A_p \cup B_p,$$

this proves $v(R_p : E_p) = v(T_p)$. The conductor is an E_p-ideal, so $\mathrm{tr}_{R_p}(E_p) = E_p(R_p : E_p) = R_p : E_p$.

Relative to Γ_p, the common ideal omits exactly $\{0\} \cup (5s + H_p)$, giving colength $p + 1$. The endomorphism formula gives exactly $p + 1$ values in $\Lambda_p \setminus \Gamma_p$, proving the balanced identity. $\square$

The equality of the two traces is consistent with the reflexive-trace criterion of Lindo–Maitra–Zhang [8]; over the present Gorenstein base it is not by itself discriminatory. The equality of the two lengths is also general structure, not a new family-specific mechanism. Indeed, applying $\mathrm{Hom}_{R_p}(-, R_p)$ to $0 \to R_p \to E_p \to E_p/R_p \to 0$ and using local duality gives

$$\ell_{R_p}(R_p/(R_p : E_p)) = \ell_{R_p}(E_p/R_p),$$

as in Herzog–Kumashiro, Proposition 3.1, Claim 1 [9]. The new family-specific data in Theorem 12.1 are the exact common ideal and the evaluation of both lengths as $p + 1$.

The first declared shorthand used the tail $[9s, \infty)$ and was refuted at $13s - 1$ by the $p = 4$ smoke check **[MV]**. That point is the Frobenius gap of Γ_p, so no R_p-ideal can contain it. The corrected formula above was then checked by two exact routes for all $p = 4, \dots, 300$, with independent reconstruction at six parameters and three rejected corruptions. The campaign and audit aggregates are

`77448398a26b958c66818b7ac4aaa4b542bad11cdba5ec7c8f7fe76db37526e2`

and

`d55ed876d7918cb4c46d8e5f3894a508d172693bde4b4c76cb67c42fcfbf0ac1`.

13 Exact conductor-stability defect

Dey proves that for a finite birational extension $R \subseteq S$ of a one-dimensional Gorenstein local ring, the conductor $R : S$ is stable if and only if S is Gorenstein [10, Corollary 3.7]. Theorem 11.1 gives $\mathrm{type}(E_p) = 10p > 1$, so this criterion already shows that $T_p = R_p : E_p$ is nonstable. The following computation quantifies that failure.

Theorem 13.1 (Exact stability defect, [D]). *For every integer $p \geq 4$,*

$$v(T_p^2) = (8s + C_p) \cup [9s, 13s - 2] \cup [13s, \infty).$$

Moreover,

$$\begin{aligned} v(T_p^2) \setminus v(t^{4s}T_p) = 8s &+ \big([p + 1, 2p] \cup [4p - 1, 5p - 2]\big) \\ &\cup 9s + \big(\{2p - 1, 4p - 1\} \cup [4p + 1, 5p - 2]\big) \\ &\cup 10s + \big([0, p] \cup \{2p - 1\} \cup [3p, 4p - 1] \cup [4p + 1, 5p - 2]\big) \\ &\cup 11s + [0, s - 1] \\ &\cup 12s + \big([2p + 1, 3p - 1] \cup [5p - 1, s - 2]\big) \\ &\cup \{17s - 1\}. \end{aligned}$$

Consequently,

$$\ell_{R_p}(T_p^2/t^{4s}T_p) = 14p.$$

Proof. Put $D = [0, s - 1]$, $D^- = [0, s - 2]$, and $U_p = A_p \cup B_p$. For residue sets $X, Y \subseteq D$, write $L(X, Y)$ for sums below s and $H(X, Y)$ for sums at least s, translated by $-s$. Expanding the defining sets into integer intervals and using $[a,b]+[c,d]=[a+c,b+d]$ gives

$$\begin{aligned} L(A_p, A_p) &= C_p, \\ H(A_p, A_p) \cup L(A_p, U_p) &= D, \\ H(A_p, U_p) \cup L(A_p, B_p) \cup L(U_p, U_p) &= D, \\ H(A_p, B_p) \cup H(U_p, U_p) \cup L(U_p, B_p) &= D, \\ H(U_p, B_p) \cup L(B_p, B_p) \cup L(A_p, C_p) &= D^-. \end{aligned}$$

These five identities list all contributions, with carries, to levels 8 through 12 of T_p^2. They give $(8s+C_p)\cup[9s,13s-2]$ and exclude the inherited gap $13s-1$. Multiplication by the least value $4s$ covers $[13s,17s-2]\cup[17s,\infty)$, while

$$17s-1=(4s+1)+(13s-2)$$

fills the intervening endpoint. This proves the square formula.

Shifting Theorem 12.1 by $4s$ and subtracting its blocks from the square gives the displayed defect. The six block cardinalities are $2p, p, 3p, 6p, 2p-1, 1$, whose sum is $14p$. □

The first EXP-015 prediction incorrectly replaced the two tail intervals by $[9s,\infty)$. The $p=4$ smoke gate refuted it at $13s-1$ before a campaign artifact was written **[MV]**. EXP-016 retained that gap, proved the corrected formula above, and checked two exact routes for all $p=4,\dots,300$. Its campaign and independent-audit aggregates are

`8c280bf1db017f678896d473461ae245b4fce57d680be94094b0884556d91853`

and

`ce4ded03761eb44cba766eafb46926c8ca83d0f7f49a26697afed5ed49905653`.

14 Conductor reduction number and Hilbert data

For an ideal I, a subideal Q is a *reduction* when $I^{r+1}=QI^r$ for some r; the least such r is the reduction number with respect to Q [11].

Theorem 14.1 (Exact conductor reduction sequence, **[D]**). *For every integer $p\ge 4$, put $Q_p = t^{4s}R_p \subset T_p$. Then Q_p is a minimal reduction of T_p with reduction number four. More precisely,*

n	0	1	2	3	$n\ge 4$
$\ell_{R_p}(T_p^{n+1}/Q_pT_p^n)$	$23p-1$	$14p$	$2p$	1	0.

The Hilbert–Samuel function satisfies

$$\ell_{R_p}(R_p/T_p^n)=24pn-39p \qquad (n\ge 4),$$

and consequently $e_0(T_p)=24p$ and $e_1(T_p)=39p$.

Proof. Theorem 13.1 supplies the square. The residue identity $\mathrm{low}(A_p+C_p)=[0,s-2]$ then gives

$$v(T_p^3)=[12s,13s-2]\cup[13s,\infty).$$

Adding $4s+A_p$ fills $[16s, 17s-2]$ and the tail from $17s$; because $1 \in A_p$, $(4s+1)+(13s-2) = 17s-1$ fills the intervening endpoint. Hence

$$v(T_p^4) = [16s, \infty), \qquad v(T_p^5) = [20s, \infty) = v(Q_p T_p^4).$$

Multiplication proves the same equality for every later power. Subtracting the shifted square and cubic profiles gives

$$\begin{aligned} v(T_p^3) \setminus v(Q_p T_p^2) &= 12s + \big([2p+1, 3p-1] \cup [5p-1, 6p-2]\big) \cup \{17s-1\}, \\ v(T_p^4) \setminus v(Q_p T_p^3) &= \{17s-1\}. \end{aligned}$$

Their cardinalities are $2p$ and 1. Theorem 13.1 gives $14p$ at the preceding stage. Finally, subtracting $v(Q_p) = 4s + \Gamma_p$ from Theorem 12.1 leaves eight disjoint blocks of sizes

$$2p-1, \quad 5p, \quad 3p, \quad 2p, \quad 3p, \quad 6p, \quad 2p-1, \quad 1,$$

whose sum is $23p-1$. The strict nonzero quotients and terminal equality prove reduction number four. Since T_p is maximal-ideal primary in dimension one, the one-generated reduction is minimal.

The parameter ideal Q_p has colength $4s = 24p$. Multiplication by its nonzerodivisor generator gives

$$\ell(T_p^i / T_p^{i+1}) = 24p - \ell(T_p^{i+1} / Q_p T_p^i).$$

Summing through $i = n-1$ and using $(23p-1) + 14p + 2p + 1 = 39p$ proves the Hilbert formula and coefficients. □

Two exact representations confirm every displayed set and length for all $p = 4, \ldots, 300$ **[MV]**. A separate tail-set implementation rehashes all 297 rows, reconstructs six parameters, and rejects deleted-terminal, false-stabilization, altered-interval, and perturbed-length controls. The campaign and audit aggregates are

`e9c3c887648f08cf67c614b381f00c8c6520dcd1bb89f8cdece62293bfd06030`

and

`0f6ed70676ffb8972b8b167ad52c4f9d2851f69c3b1d96f4023e5e3d5825c781`.

15 The depth-zero conductor tangent cone

Let

$$G_p = \mathrm{gr}_{T_p}(R_p) = \bigoplus_{n\geq 0} T_p^n/T_p^{n+1}$$

and retain the minimal reduction $Q_p = (x)$, where $x = t^{4s}$. The Valabrega–Valla criterion detects regularity of the initial form x^* through the intersections $Q_p \cap T_p^{n+1}$ [12].

Theorem 15.1 (Exact tangent-cone defect, **[D]**)**.** *For every integer $p \geq 4$, put*

$$K_p = \{2p-1, 4p-1\} \cup [4p+1, 5p-2].$$

Then the complete Valabrega–Valla module is concentrated in degree one:

$$v\left(\frac{Q_p \cap T_p^2}{Q_pT_p}\right) = 9s + K_p, \qquad \ell_{R_p}\left(\frac{Q_p \cap T_p^2}{Q_pT_p}\right) = p,$$

while

$$Q_p \cap T_p^{n+1} = Q_pT_p^n \qquad (n = 0 \text{ and every } n \geq 2).$$

Consequently

$$\dim G_p = 1, \qquad \operatorname{depth} G_p = 0,$$

so G_p is not Cohen–Macaulay. Its Hilbert series is

$$H_{G_p}(z) = \frac{(p+1) + (9p-1)z + 12pz^2 + (2p-1)z^3 + z^4}{1-z}.$$

In particular every coefficient of the numerator is positive.

Proof. Write $D = [0, s-1]$ and $U = A_p \cup B_p$. Directly from the residue definitions,

$$D \setminus U = K_p, \qquad |K_p| = p.$$

The exact profiles from Theorems 12.1 and 13.1 give

$$\begin{aligned}
v(Q_p) &= \{4s\} \cup (8s + A_p) \cup (9s + D) \cup (10s + B_p) \cup (12s + C_p) \cup [13s, \infty),\\
v(Q_pT_p) &= (8s + A_p) \cup (9s + U) \cup (10s + B_p) \cup (12s + C_p) \cup [13s, \infty),\\
v(T_p^2) &= (8s + C_p) \cup ([9s, \infty) \setminus \{13s-1\}).
\end{aligned}$$

Since $A_p \subset C_p$ and $C_p \subset [0, s-2]$, intersection leaves exactly the additional block $9s + (D \setminus U) = 9s + K_p$ beyond $v(Q_pT_p)$. This proves the first formula and its cardinality.

Theorem 14.1 gives

$$v(T_p^3) = [12s, \infty) \setminus \{13s-1\}, \quad v(T_p^4) = [16s, \infty), \quad v(T_p^5) = [20s, \infty).$$

Intersecting each with $v(Q_p)$ gives respectively $v(Q_pT_p^2)$, $v(Q_pT_p^3)$, and $v(Q_pT_p^4)$. For every $n \geq 4$, stabilization $T_p^{n+1} = Q_pT_p^n \subset Q_p$ proves the intersection equality immediately. The case $n = 0$ is $Q_p \cap T_p = Q_p$.

The ring R_p is a one-dimensional Cohen–Macaulay local domain with infinite residue field, x is regular, and Q_p is a minimal reduction. Hence x^* is a homogeneous parameter of G_p. The nonzero intersection defect makes x^* a zero divisor by the Valabrega–Valla criterion, so the one-dimensional ring G_p has depth zero.

Finally write

$$a_n = \ell(T_p^{n+1}/Q_pT_p^n).$$

Theorem 14.1 gives

$$(a_0, a_1, a_2, a_3, a_4, \ldots) = (23p-1, 14p, 2p, 1, 0, \ldots).$$

Since $\ell(T_p^n/xT_p^n) = \ell(R_p/xR_p) = 24p$, the natural exact sequence gives

$$h_n := \ell(T_p^n/T_p^{n+1}) = 24p - a_n.$$

Thus $(h_0, h_1, h_2, h_3, h_4, \ldots) = (p+1, 10p, 22p, 24p-1, 24p, \ldots)$, and multiplication of $\sum h_n z^n$ by $1-z$ gives the displayed numerator. □

Two exact implementations verify every intersection, length, and Hilbert coefficient for all $p = 4, \ldots, 300$ [**MV**]. A separately written bounded-bitset audit reconstructs six parameters and rehashes all 297 campaign rows. Deleted-witness, false-later-defect, false-Cohen–Macaulay, and perturbed-numerator controls are rejected. The campaign and audit aggregates are

`9631c644732f0921be3b3027e18a01110f23dad897fbf2cb14dd3a493eda5971`

and

`7c2abcd290bc3461fc5251bc3372e20a7fa25c888e3b2ed635368b4dda0781ff`.

15.1 Complete graded torsion and Buchsbaumness

Let $\mathcal{M}_p = (\mathfrak{m}_p/T_p) \oplus G_{p,+}$ be the homogeneous maximal ideal of G_p. The dimension-one local-cohomology criterion says that G_p is Buchsbaum precisely when $\mathcal{M}_p H^0_{\mathcal{M}_p}(G_p) = 0$ [16, 17]. The cited semigroup formulas concern maximal-ideal filtrations; here we compute the local cohomology directly for the conductor filtration.

Theorem 15.2 (Buchsbaum tangent cones with unbounded invariant, [**D**])**.** *For every integer $p \geq 4$, the complete zeroth local cohomology is concentrated in degree zero:*

$$H^0_{\mathcal{M}_p}(G_p) = \operatorname{span}_{\Bbbk}\{t^{5s+h} + T_p : h \in K_p\} \cong \Bbbk^p.$$

Moreover

$$\mathcal{M}_p H^0_{\mathcal{M}_p}(G_p) = 0.$$

Consequently G_p is Buchsbaum but not Cohen–Macaulay, and its Buchsbaum invariant is

$$I(G_p) = \ell(H^0_{\mathcal{M}_p}(G_p)) = p.$$

The quotient $G_p/H^0_{\mathcal{M}_p}(G_p)$ is Cohen–Macaulay and has Hilbert series

$$\frac{1 + (10p-1)z + 12pz^2 + (2p-1)z^3 + z^4}{1-z}.$$

Proof. A homogeneous class represented by $a \in T_p^n$ lies in zeroth local cohomology exactly when $aT_p^k \subseteq T_p^{n+k+1}$ for some k. Thus

$$H^0_{\mathcal{M}_p}(G_p)_n = \bigcup_{k\geq 1} \frac{(T_p^{n+k+1} : T_p^k) \cap T_p^n}{T_p^{n+1}}.$$

Here saturation by $G_{p,+}$ and by $\mathcal{M}_p$ agree because the maximal ideal of the Artinian degree-zero ring R_p/T_p is nilpotent.

Theorem 14.1 gives $v(T_p^k) = [4ks, \infty)$ for every $k \geq 4$. Therefore, for a monomial t^v and any $k \geq 4$,

$$t^v T_p^k \subseteq T_p^{n+k+1} \iff v \geq 4(n+1)s.$$

Any containment at a smaller power persists after multiplication to a power at least four, so this stable threshold is the complete colon saturation.

The exact profiles give

$$v(R_p) \setminus v(T_p) = \{0\} \cup (5s + K_p).$$

The degree-zero threshold $4s$ deletes the unit and retains $5s + K_p$. In degrees one, two, and three, the consecutive power profiles agree from the respective thresholds $8s$, $12s$, and $16s$ onward. For every $n \geq 4$, stabilization makes them agree from $4(n+1)s$ onward. Hence the displayed p degree-zero classes are the complete local cohomology.

It remains to test the full homogeneous maximal ideal. If $r = 5s + h$ with $h \in K_p$, every product of t^r with $\mathfrak{m}_p$ or T_p has value at least $9s$. Both T_p and T_p^2 contain all such values except $13s - 1$. Equality $r + a = 13s - 1$ would force $a = 8s - 1 - h$, strictly between levels $7s$ and $8s$, but the exact ring profile has no value there. Thus

$$t^r \mathfrak{m}_p \subseteq T_p, \qquad t^r T_p \subseteq T_p^2.$$

The degree-zero and positive parts of $\mathcal{M}_p$ both annihilate every torsion class. The dimension-one criterion proves Buchsbaumness; Theorem 15.1 proves depth zero. The Buchsbaum invariant is therefore the displayed length p.

Quotienting by the full zeroth local cohomology leaves a one-dimensional ring of positive depth, hence a Cohen–Macaulay ring. Since the torsion Hilbert series is the constant p, subtracting it from Theorem 15.1's Hilbert series gives the asserted numerator. □

Two exact implementations verify the full colon saturation and both maximal-ideal actions for all $p = 4, \ldots, 300$ [**MV**]. An independently written bounded-bitset audit reconstructs six parameters and rehashes all 297 rows. Unit, deleted-torsion, false-positive-degree, degree-zero-action, false-invariant, and perturbed-series controls are rejected. The campaign and audit aggregates are

`854d7889d9d7b911b462e4d483e021210ae2873ae0ec0091ec30e8fb29d6dbf7`

and

`0b01853febc9e9754e28abcd099a7ae3a97f4cc0ab92f3a345ab2ae03cd3c68a`.

15.2 The complete module over the Noether normalization

Put $F_p = \Bbbk[x_p]$, where $x_p = (t^{4s})^*$ has degree one. The reduction theorem makes $F_p \to G_p$ a Noether normalization. This viewpoint follows the graded-module framework of Cortadellas Benitez and Zarzuela [13, 15], while the conductor-filtration calculation below is derived directly from Theorems 14.1, 15.1, and 15.2.

Theorem 15.3 (Exact Noether-normalization module, [**D**])**.** *For every integer $p \geq 4$, there is a graded F_p-module isomorphism*

$$\begin{aligned} G_p \cong (F_p/(x_p))^p \oplus F_p \oplus F_p(-1)^{10p-1} \oplus F_p(-2)^{12p} \\ \oplus F_p(-3)^{2p-1} \oplus F_p(-4). \end{aligned}$$

Its minimal graded F_p-resolution is

$$0 \longrightarrow F_p(-1)^p \longrightarrow F_p^{p+1} \oplus F_p(-1)^{10p-1} \oplus F_p(-2)^{12p} \oplus F_p(-3)^{2p-1} \oplus F_p(-4) \longrightarrow G_p \longrightarrow 0.$$

Consequently

$$\operatorname{pd}_{F_p}(G_p) = 1, \qquad \operatorname{reg}_{F_p}(G_p) = 4, \qquad a(G_p) = 3,$$

and

$$\ell(G_p/x_pG_p) = 25p = e_0(T_p) + I(G_p).$$

Proof. Theorem 14.1 gives $T_p^{n+1} = t^{4s}T_p^n$ for $n \geq 4$, so G_p is finite over F_p. Since G_p/x_pG_p has finite length, F_p-torsion equals zeroth local cohomology. Theorem 15.2 therefore gives

$$\operatorname{tors}_{F_p}(G_p) = H^0_{\mathcal{M}_p}(G_p) \cong (F_p/(x_p))^p,$$

with all generators in degree zero. In particular, every torsion elementary divisor has exponent one.

The quotient $C_p = G_p/H^0_{\mathcal{M}_p}(G_p)$ is finite and torsion-free over the principal ideal domain F_p, hence graded free. Its Hilbert series from Theorem 15.2 is

$$H_{C_p}(z) = \frac{1 + (10p-1)z + 12pz^2 + (2p-1)z^3 + z^4}{1-z}.$$

Since $H_{F_p(-d)}(z) = z^d/(1-z)$, the numerator uniquely gives the five displayed free shifts. The quotient is projective, so the exact sequence with the torsion submodule splits after choosing homogeneous lifts of a free basis. This proves the decomposition.

Resolving each $F_p/(x_p)$ by multiplication by x_p gives the displayed minimal resolution. Its only first Betti number is $\beta_{1,1} = p$; its degree-zero through degree-four generator counts are $p+1, 10p-1, 12p, 2p-1, 1$. Therefore the projective dimension is one and the resolution formula for regularity gives four. Finite-length torsion has no first local cohomology, while the largest free shift is four and end $H^1(F_p(-d)) = d-1$; hence $a(G_p) = 3$.

Finally, modulo x_p every cyclic summand contributes one $\Bbbk$-dimension. The free rank is

$$1 + (10p-1) + 12p + (2p-1) + 1 = 24p = e_0(T_p),$$

and the torsion contributes $p = I(G_p)$. This proves the parameter-section identity. □

For an independent exact reconstruction, EXP-020 forms the Apery table of $R_p, T_p, \ldots, T_p^5$ modulo $24p$. Each column splits into finite and infinite strings under multiplication by x_p, recovering every cyclic summand. The recursive-power and Hilbert/torsion routes agree for all $p = 4, \ldots, 300$ [**MV**]; a separate closed-form implementation rebuilds six parameters and rehashes all 297 rows. Exponent-two-torsion, deleted-free-summand, perturbed-Betti, false-regularity, and false-section-length controls are rejected. The campaign and audit aggregates are

`02cf6f62a71de1a897cd46149e8c89d1c55bf810d28dddc02fc6c5330b9c1aed`

and

`c439f7e4fbd3cee983f32e5c6a27b347017c165cdf6d9fee54eb8d53ab634eac`.

15.3 The conductor special fiber

Let

$$\mathcal{F}_p = \mathcal{F}(T_p) = \bigoplus_{n\geq 0} T_p^n/\mathfrak{m}_p T_p^n$$

be the special fiber. There is a natural graded-algebra surjection $G_p \to \mathcal{F}_p$ because $T_p^{n+1} \subseteq \mathfrak{m}_p T_p^n$.

Theorem 15.4 (Canonical fiber cone and exact type, **[D]**)**.** *For every integer $p \geq 4$,*

$$T_p^2 = \mathfrak{m}_p T_p, \qquad T_p^{n+1} = \mathfrak{m}_p T_p^n \quad (n \geq 1),$$

and the natural map induces a graded-algebra isomorphism

$$G_p/H^0_{\mathcal{M}_p}(G_p) \cong \mathcal{F}_p.$$

The ring $\mathcal{F}_p$ is one-dimensional Cohen–Macaulay, with

$$\begin{aligned}\mathcal{F}_p \cong F_p \oplus F_p(-1)^{10p-1} \oplus F_p(-2)^{12p} \\ \oplus F_p(-3)^{2p-1} \oplus F_p(-4)\end{aligned}$$

as a graded F_p-module. Its multiplicity is $24p$, its reduction number and regularity are four, and $a(\mathcal{F}_p) = 3$. The Artinian reduction $B_p = \mathcal{F}_p/x_p\mathcal{F}_p$ has

$$h(B_p) = (1, 10p-1, 12p, 2p-1, 1)$$

and graded socle dimensions

$$(0, 0, 10p, 0, 1).$$

Thus $\mathrm{type}(\mathcal{F}_p) = 10p + 1$*, and $\mathcal{F}_p$ is neither level nor Gorenstein.*

Proof. Minkowski addition of the exact value blocks in Theorems 12.1 and 13.1 gives

$$v(\mathfrak{m}_p T_p) = (8s + C_p) \cup [9s, 13s-2] \cup [13s, \infty) = v(T_p^2).$$

Indeed, the lowest block follows from $A_p + A_p = C_p \cup [6p, 8p-4]$; the latter interval and all remaining block sums lie at level at least $9s$ and fill every ring value there. The Frobenius value $13s-1$ remains absent. Equality of the monomial value sets proves $T_p^2 = \mathfrak{m}_p T_p$, and multiplication by T_p^{n-1} proves the higher identities.

The degree-zero kernel of $G_p \to \mathcal{F}_p$ is $\mathfrak{m}_p/T_p$, which is exactly Theorem 15.2's H^0. In degree $n \geq 1$, the kernel is $\mathfrak{m}_p T_p^n/T_p^{n+1} = 0$. This proves the natural algebra isomorphism. Cohen–Macaulayness and the displayed free module now follow from Theorems 15.2 and 15.3; the Hilbert, multiplicity, reduction, regularity, and a-invariant claims follow immediately.

For the type, reduce modulo the regular element x_p. Direct subtraction of $\mathfrak{m}_p T_p^n + t^{4s} T_p^{n-1}$ from the exact power blocks gives the displayed h-vector. The degree-two nonsocle offsets are precisely

$$[p+1, 2p] \cup [4p-1, 5p-2].$$

They multiply into the two degree-three blocks by degree-one offsets in $[1, p]$. Every other degree-two basis offset kills degree one. No degree-three basis element is socle: if its offset is d, the complementary offset $6p - 1 - d$ belongs to the degree-one basis and their product is the unique degree-four class. That class is socle because degree five vanishes. Hence the only socle dimensions are $10p$ in degree two and one in degree four, proving the final claims. □

EXP-021 reconstructs the square, natural kernel, Artinian basis, and socle for all $p = 4, \dots, 300$ [**MV**]. The first full attempt passed through $p = 225$ but exceeded its declared budget and is preserved as inconclusive. An exact bitset implementation then completed all 297 parameters, and an independently encoded audit rebuilt six parameters and rehashed every row. The campaign and audit aggregates are

`3857877586143a3be5f14852feb12bd9efbfdf7c1cde458f30e8cd689155a95b`

and

`1779407050b199039d3f6d808a720ea051a81ef11734fd1bccd1a76ec78c0a9c`.

15.4 The defining ideal of the special fiber

We now refine Theorem 15.4 from a module decomposition to a minimal algebra presentation. All intervals in this subsection are integer intervals. Let $\mathcal{E}_{p,d}$ be the offset basis of $(\mathcal{F}_p)_d$, after subtracting the base value $4sd$. The exact power calculation gives

$$\begin{aligned}\mathcal{E}_{p,1} &= [0, p] \cup [3p, 4p-2] \cup [6p, 8p-2] \cup [8p, 10p-2] \cup \{10p\} \\ &\quad \cup [11p-1, 12p-1] \cup [13p+1, 14p-2] \cup [14p, 15p-1] \cup \{16p\} \cup [17p-1, 18p-1], \qquad (5)\\ \mathcal{E}_{p,2} &= [0, 2p] \cup [3p, 5p-2] \cup [6p, 24p-1], \\ \mathcal{E}_{p,3} &= [0, 24p-1] \setminus \{6p-1\}, \qquad \mathcal{E}_{p,4} = \mathcal{E}_{p,5} = [0, 24p-1]. \qquad (6)\end{aligned}$$

In particular, $|\mathcal{E}_{p,1}| = 10p$. Set

$$P_p = \Bbbk[X_a : a \in \mathcal{E}_{p,1}]$$

and let $\varphi_p : P_p \twoheadrightarrow \mathcal{F}_p$ send X_a to the class of t^{4s+a}. Write $\mathfrak{a}_p = \ker \varphi_p$.

Theorem 15.5 (One-cubic defining ideal, [**MV**])**.** *For every integer $p \geq 4$,*

$$\mathfrak{a}_p = ((\mathfrak{a}_p)_2, f_p), \qquad f_p = X_0^2 X_{3p} - X_p^3.$$

Equivalently, the first graded Betti row over P_p is

$$\beta_{1,2}^{P_p}(\mathcal{F}_p) = 50p^2 - 17p, \qquad \beta_{1,3}^{P_p}(\mathcal{F}_p) = 1, \qquad \beta_{1,j}^{P_p}(\mathcal{F}_p) = 0 \quad (j \geq 4).$$

Hence $\mu(\mathfrak{a}_p) = 50p^2 - 17p + 1$, the relation type is three, and $\mathcal{F}_p$ is not Koszul.

Proof. The semigroup monomials are linearly independent. Consequently, a monomial in the variables X_a is zero in $\mathcal{F}_p$ exactly when its total offset is absent from the corresponding $\mathcal{E}_{p,d}$, and two nonzero monomials have the same image exactly when their total offsets agree. There is no linear equation. Theorem 15.4 gives $\dim_{\Bbbk}(\mathcal{F}_p)_2 = 22p$, so

$$\beta_{1,2}^{P_p}(\mathcal{F}_p) = \binom{10p+1}{2} - 22p = 50p^2 - 17p. \qquad (7)$$

The offsets $0, p, 3p$ belong to $\mathcal{E}_{p,1}$, and $0 + 0 + 3p = p + p + p$, so $f_p \in \mathfrak{a}_p$. It is not generated by quadrics. Indeed, the only unordered pair in $\mathcal{E}_{p,1}$ with sum $3p$ is $\{0, 3p\}$, the only pair with sum $2p$ is $\{p, p\}$, and the only pair with sum zero is $\{0, 0\}$. Thus neither monomial in f_p admits a nontrivial quadratic replacement, and the two cannot be joined by a chain of quadratic moves.

It remains to prove that this is the only higher equation. Suppose the complete kernel is known through degree $d-1$, where $d \in \{3,4,5\}$. For a fixed total t, form the graph

$$V_{d,t} = \{a \in \mathcal{E}_{p,1} : t-a \in \mathcal{E}_{p,d-1}\}.$$

Two vertices a, c are adjacent when

$$t-a-c \in \mathcal{E}_{p,d-2}.$$

This is exactly the commutative factorization move between X_a times the preceding class of offset $t-a$ and X_c times the class of offset $t-c$. Add a zero vertex when the same factorization makes the alternate degree-$(d-1)$ remainder invalid. Therefore the degree-d quotient by all lower equations has one basis element for each nonzero graph component. It follows that

$$\beta^{P_p}_{1,d}(\mathcal{F}_p) = \#\{\text{nonzero state components in degree } d\} - |\mathcal{E}_{p,d}|. \tag{8}$$

For degree three, the exact component reduction is

Total range	common hub	path length at most
$[0, 2p]$	0	3
$[2p+1, 3p-1]$	$t-2p$	3
$[3p+1, 5p-2]$	0	3
$[5p-1, 6p-2]$	$t-(5p-2)$	3
$[6p, 24p-1]$	0	3

Every invalid total reaches the zero vertex in at most one move. At the omitted total $t = 3p$, the vertices are exactly $0, p, 3p$; the vertices 0 and $3p$ are joined, while p is isolated. Hence there are exactly two components, represented by the two monomials in f_p. Adjoining f_p joins them.

For degree four, zero is a hub except at $t = 6p-1$, where p is a hub. To see the only nontrivial reduction, put

$$H = [2p+1, 3p-1] \cup [5p-1, 6p-2] = [0, 6p-2] \setminus \mathcal{E}_{p,2}.$$

If $t-a \in \mathcal{E}_{p,2}$, the state is adjacent to zero. If $t-a \in H$, it moves first to p, then either to zero or through $t-3p$, which belongs to $[1, p-1]$ or $[3p, 4p-2]$. The apparent boundary $t = 7p-1$, where $t-p$ would be the hole in $\mathcal{E}_{p,3}$, cannot occur with both $t-a \in H$ and $a \in \mathcal{E}_{p,1}$. At $t = 6p-1$, the vertex set is $[1, p] \cup [3p, 4p-2]$, connected through p. Invalid high totals die directly. Thus the degree-four defect is zero.

For degree five, $\mathcal{E}_{p,4}$ is full and $\mathcal{E}_{p,3}$ misses only $h = 6p-1$. A state has a direct edge to zero unless $t-a = h$. In the exceptional case, moving through the generator 1, or through 2 when $a = 1$, gives a state with a direct zero edge. Hence the degree-five defect is also zero.

The degree-three table and every boundary assertion in the degree-four and degree-five reductions were encoded as negated Presburger formulas over all integers $p \geq 4$ [**MV**]. The exact solver closed all 133 terminal formulas as UNSAT, after four affine-cell bisections, with no SAT or unresolved result. A separately encoded total-by-total graph rehashed the complete campaign and rebuilt three spaced parameters. Thus (8) gives one cubic and no new equation in degrees four or five.

Finally, T_p has reduction number four by Theorem 14.1, and $\mathcal{F}_p$ is Cohen–Macaulay by Theorem 15.4. The defining-ideal construction of Abdolmaleki–Kumashiro [14, Theorem 2.8] therefore supplies a complete set of equations through degree five. No higher minimal equation can occur. This proves the Betti row and relation type. A standard graded Koszul algebra has a quadratic defining ideal, while f_p is a necessary minimal cubic, so $\mathcal{F}_p$ is not Koszul. □

The bounded state-graph campaign passes every $p = 4, \dots, 23$ **[MV]**. Its first requested extension through $p = 24$ completed that row but crossed the declared five-minute budget, and is preserved as inconclusive. The successful campaign, independent audit, and Presburger-query aggregates are respectively

`d23792c47a2e07785a27ebc71e99619705f7aa53a38ebe7f66ffa03b0518ce83`,

`a27b3b13fde197b1f011bf07dc2c321d84ab7c895c9aa02d7c2a073e48f18038`,

and

`832c8421fe66359b8c246e3465e27de6ea7829215f892ab815e72b1f44787194`.

15.5 Homological edges over the presentation ring

The h-vector and socle in Theorem 15.4, together with the complete first Betti row in Theorem 15.5, force more of the minimal resolution than the defining equations alone suggest. Put

$$N = 10p, \qquad c = N - 1 = 10p - 1,$$

and write $\beta_{i,j} = \beta_{i,j}^{P_p}(\mathcal{F}_p)$.

Theorem 15.6 (Homological edge data, **[D]**)**.** *For every integer $p \geq 4$,*

$$\mathrm{pd}_{P_p}(\mathcal{F}_p) = c, \qquad \mathrm{reg}_{P_p}(\mathcal{F}_p) = 4,$$

and

$$\sum_{i,j} (-1)^i \beta_{i,j} z^j = (1 - z)^c \big(1 + (10p - 1)z + 12pz^2 + (2p - 1)z^3 + z^4\big). \tag{9}$$

The first linear-syzygy count is

$$\beta_{2,3} = \frac{2p(500p^2 - 330p + 31)}{3}. \tag{10}$$

The complete last homological row and a penultimate extremal entry are

$$\beta_{c,c+2} = 10p, \qquad \beta_{c,c+4} = 1, \qquad \beta_{c,j} = 0 \ (j \notin \{c + 2, c + 4\}), \qquad \beta_{c-1,c+3} = 8p. \tag{11}$$

Finally, the canonical module $\omega_{\mathcal{F}_p}$ has $10p$ minimal generators in degree -1 and one in degree -3.

Proof. The polynomial ring P_p has depth N, whereas Theorem 15.4 gives $\mathrm{depth}_{P_p}(\mathcal{F}_p) = \dim \mathcal{F}_p = 1$. Auslander–Buchsbaum therefore gives projective dimension $N - 1 = c$. The same theorem gives the h-polynomial

$$h_p(z) = 1 + (10p - 1)z + 12pz^2 + (2p - 1)z^3 + z^4.$$

Since $\mathcal{F}_p$ is Cohen–Macaulay, its regularity is $\deg h_p = 4$. In particular, regularity over the full presentation ring is already determined, rather than a separate open question. Taking Hilbert series in a minimal P_p-resolution gives

$$\frac{h_p(z)}{1 - z} = \frac{\sum_{i,j} (-1)^i \beta_{i,j} z^j}{(1 - z)^N},$$

which proves (9).

There is no linear equation, so minimality gives no shift below $i+1$ in homological degree i. Consequently the coefficient of z^3 in (9) is $\beta_{2,3} - \beta_{1,3}$. Theorem 15.5 gives $\beta_{1,3} = 1$, and expansion yields

$$\beta_{2,3} = 1 - \binom{c}{3} + c\binom{c}{2} - 12pc + (2p-1) = \frac{2p(500p^2 - 330p + 31)}{3}.$$

Independently, the degree-three part of the first free module has dimension $N\beta_{1,2} + \beta_{1,3}$ and maps onto $(\mathfrak{a}_p)_3$. Since

$$\dim(P_p)_3 = \binom{N+2}{3}, \qquad \dim(\mathcal{F}_p)_3 = 24p - 1,$$

its kernel has dimension

$$N(50p^2 - 17p) + 1 - \left(\binom{N+2}{3} - (24p-1)\right),$$

which simplifies to the same value.

Now reduce modulo X_0, whose image x_p is the regular parameter from Theorem 15.4. Because X_0 is regular on both P_p and $\mathcal{F}_p$, tensoring the minimal resolution with $Q_p = P_p/(X_0)$ remains exact and minimal. Hence its Betti numbers equal those of the Artinian reduction $B_p = \mathcal{F}_p/x_p\mathcal{F}_p$ over the polynomial ring Q_p in c variables. Top Koszul homology identifies

$$\operatorname{Tor}_c^{Q_p}(B_p, \Bbbk)_j \cong \operatorname{Soc}(B_p)_{j-c}.$$

Theorem 15.4 gives socle dimensions $10p$ in degree two, one in degree four, and zero otherwise. This proves the complete last row in (11).

In internal degree $c+3$, regularity four kills every contribution in homological degree at most $c-2$, while the last-row calculation kills $\beta_{c,c+3}$. Thus the coefficient of z^{c+3} in (9) is $(-1)^{c-1}\beta_{c-1,c+3}$. Only the last two terms of h_p contribute, giving

$$(-1)^c(2p-1) + (-1)^{c-1}c = (-1)^{c-1}8p.$$

Therefore $\beta_{c-1,c+3} = 8p$.

Finally,

$$\omega_{\mathcal{F}_p} = \operatorname{Ext}^c_{P_p}(\mathcal{F}_p, P_p(-N)).$$

Minimality lets us read its minimal generators from the dual of the last free module. A summand $P_p(-j)$ contributes in degree $N-j$. The $10p$ last summands with $j = c+2 = N+1$ therefore give degree -1, and the single summand with $j = c+4 = N+3$ gives degree -3. □

EXP-024 checks both derivations of (10), every selected coefficient in (9), and the socle-to-last-row transfer for all $p = 4, \ldots, 300$ **[MV]**. Its independent implementation rebuilds all 297 rows, reconstructs six spaced EXP-021 socle cases, cross-checks all twenty persisted EXP-023 first rows, and rejects seven adversarial corruptions. The campaign and audit aggregates are

`baf6200a442be9476cd083fde753bbdd9e623c06aa2528f3a7f138ee825637eb`

and

`b6035f615f2b2092351b5a42e5a734c72ba4783adf82943ed41b38fe07ef17e2`.

The symbolic argument above, not the finite campaign, proves the theorem. Equation (9) is alternating data and does not determine the unresolved interior of the Betti table.

15.6 The first interior Betti strand

We retain the offset in (5) by declaring $\deg X_a = (1, a)$. Write $\beta_{i,(j,b)}$ for the resulting bigraded Betti number and $\beta_{i,j} = \sum_b \beta_{i,(j,b)}$. Let

$$\mathfrak{q}_p = ((\mathfrak{a}_p)_2), \qquad \mathfrak{a}_p = (\mathfrak{q}_p, f_p)$$

as in Theorem 15.5.

Theorem 15.7 (First interior Betti strand, **[D]**)**.** *For every integer $p \geq 4$ and every field $\Bbbk$,*

$$\beta_{2,4} = 8p, \qquad \beta_{3,4} = \frac{p(5p-1)(500p^2 - 440p + 47)}{2}. \tag{12}$$

Moreover, $\beta_{2,(4,b)} = 1$ exactly for

$$\begin{aligned} \mathcal{S}_p = {} & [9p, 11p-2] \cup [11p, 13p-2] \cup \{13p\} \cup [14p-1, 15p-1] \\ & \cup [16p+1, 17p-2] \cup [17p, 18p-1] \cup \{19p\} \cup [20p-1, 21p-1], \end{aligned} \tag{13}$$

and it is zero at every other offset. Equivalently,

$$\mathcal{S}_p = \{3p + a : a \in \mathcal{E}_{p,1},\ a \geq 6p\}.$$

The corresponding integral relative homology is free of rank one at every supported offset; in particular, the result has no characteristic exception.

Proof. We first identify the exact finite chain complexes. Put

$$H_p = \langle (1, a) : a \in \mathcal{E}_{p,1} \rangle \subseteq \mathbb{N}^2, \qquad M_p = (t^{(n,u)} : u \geq 24p) \subseteq \mathbb{Z}[H_p].$$

The offset calculation (6) identifies the special fiber with the base change of $\mathbb{Z}[H_p]/M_p$. For a bidegree $w = (j, b)$, define the squarefree divisor pair

$$\begin{aligned} \Delta_w &= \{F \subseteq \mathcal{E}_{p,1} : w - (|F|, \textstyle\sum F) \in H_p\}, \\ \Gamma_w &= \{F \in \Delta_w : t^{w - (|F|, \sum F)} \in M_p\}. \end{aligned}$$

The relative squarefree-divisor construction is standard for affine semigroup quotients by monomial ideals [18]; see also [19] for the semigroup-ring Betti framework and its characteristic warning. Here the identification is immediate: the bidegree-w Koszul chain group in homological degree r has one basis vector for every r-subset F satisfying

$$b - \sum F \in \mathcal{E}_{p,j-r},$$

and signed deletion is exactly the relative simplicial boundary. Hence

$$\beta_{i,w} = \dim_{\Bbbk} \widetilde{H}_{i-1}(\Delta_w, \Gamma_w; \Bbbk). \tag{14}$$

We next give an integral upper bound in total degree four. Order $\mathcal{E}_{p,1}$ increasingly. In every offset block, scan the vertices in this order and pair a still-unmatched cell F with $F \cup \{v\}$ whenever both survive; within a scan use increasing dimension and lexicographic order. Every pair is a unit boundary entry, and the pivot order strictly increases along a reversed matched arrow, so this is an acyclic integral matching. Substitution of (5)–(6) leaves the following complete critical-cell list in relative dimensions zero and one. In the middle row write $b = 6p + r$.

Offset	critical 0-cells	critical 1-cells
$b \in \mathcal{S}_p$	$\{0\}$	$\{p, b-3p\}$
$6p \le b \le 7p-3$	$\{0\}, \{r+2\}$	$\{p, 3p+r\}$
all other b	immaterial	none

In the middle row, the unique alternating path sends the critical edge to a unit multiple of $\{r+2\} - \{0\}$, so it contributes no homology. The table follows by scanning the eleven blocks in (5): the only toggle obstructions are the gaps of $\mathcal{E}_{p,2}$, the hole $6p-1$ of $\mathcal{E}_{p,3}$, and a gap of $\mathcal{E}_{p,1}$. All candidate edges outside the two displayed rows toggle at their first available vertex. Therefore, over every field,

$$\beta_{2,(4,b)} \le 1 \quad (b \in \mathcal{S}_p), \qquad \beta_{2,(4,b)} = 0 \quad (b \notin \mathcal{S}_p). \tag{15}$$

For the matching lower bound, we compute the linear part of the quadratic colon. Define

$$V_p = \{v \in \mathcal{E}_{p,1} : v-p, v+p \in \mathcal{E}_{p,1}\} = [7p, 8p-2] \cup [8p, 9p-2] \cup \{9p, 10p\}.$$

For every $a \in \mathcal{E}_{p,1}$ with $a \ge 6p$, the factorizations $(0,0,3p,a)$ and (p,p,p,a) are connected by quadratic moves. If $a+p, a+2p \in \mathcal{E}_{p,1}$, use

$$(0,0,3p,a) \sim (0,0,p,a+2p) \sim (0,p,p,a+p) \sim (p,p,p,a).$$

For $a = 7p-1$, use

$$(0,0,3p,7p-1) \sim (0,1,3p,7p-2) \sim (0,1,p,9p-2) \sim (1,p,p,8p-2) \sim (p,p,p,7p-1).$$

In every remaining case write $a = u+v$, with $u \in \mathcal{E}_{p,1}$ and $v \in V_p$, and use

$$(0,0,3p,a) \sim (0,3p,u,v) \sim (p,3p,u,v-p) \sim (p,p,u,v+p) \sim (p,p,p,a).$$

The block formula (5) verifies that these cases cover all high offsets.

They cover no low offset. For a factorization of total at most $7p-2$, color a generator odd exactly when it lies in $[3p, 4p-2]$. Every equality of two pair sums preserves the parity of the number of odd generators: the sole cross-block possibility replaces two middle-block generators by one low and one high generator. If $a \le p$, the two factorizations of $X_a f_p$ have parities one and zero; if $3p \le a \le 4p-2$, they have parities zero and one. Thus

$$(\mathfrak{q}_p : f_p)_1 = \operatorname{span}_{\Bbbk}\{X_a : a \in \mathcal{E}_{p,1},\ a \ge 6p\}, \tag{16}$$

which has dimension $8p$.

Apply the mapping cone to

$$0 \longrightarrow P_p/(\mathfrak{q}_p : f_p)(-3) \xrightarrow{f_p} P_p/\mathfrak{q}_p \longrightarrow \mathcal{F}_p \longrightarrow 0.$$

Every linear generator in (16) supplies a homological-degree-two summand shifted by four. No cancellation is possible: its maps to the shift-three free summand by a linear entry and to the quadratic-generator module by degree-two entries. Thus there is one nonzero class at each offset $3p+a$ in (13). Combined with (15), this proves $\beta_{2,4} = 8p$, the exact support, and freeness over $\mathbb{Z}$: the matching is integral and each mapping-cone class is primitive.

Finally, no minimal degree-four equation exists by Theorem 15.5, and minimal shifts exclude homological degree at least four from internal degree four. The coefficient of z^4 in (9) is therefore

$$\beta_{2,4} - \beta_{3,4} = -\frac{p(2500p^3 - 2700p^2 + 675p - 63)}{2}.$$

Substitution of $\beta_{2,4} = 8p$ gives the second formula in (12). □

EXP-027 checks the complete offset chain profile for $p = 4, 5, 6$, with totals $32, 40, 48$, and the smallest case agrees over GF(2) and GF(1000003) [**MV**]. The closed formulas and explicit quadratic paths pass every $p = 4, \ldots, 300$; six all-parameter counterexample queries are UNSAT. An independently written audit reconstructs the full $p = 4$ chain profile and six spaced formula rows. The campaign, symbolic, and audit aggregates are

`9bc93c41d899df4a39b85a452257d114000df84730d44f0661c19b7dd8322b63`,

`b7e760290880e821fbc2ef86b03279edab1543307fd5c58cbb635d9c4b765db8`,

and

`f30b94ce86638732a407a3bb5abb4dfde8a1258441df71a36138dd0ae129d454`.

The finite campaigns validate the implementation. The relative-chain and matching arguments, together with the colon and mapping cone, prove the theorem.

16 Reproducibility and trust boundary

All compact code, hypotheses, manifests, audits, and verdicts are in the public repository under

`problems/commutative-algebra/huneke-wiegand/`.

Heavy CNFs and proofs are retained outside Git with hashes committed in the manifests. The minimum-layer archive adds 12 fully accounted files and a reconstruction auditor. The record also includes the compact EXP-011–027 campaigns, independent reconstruction audits, symbolic proofs, the preserved EXP-013 correction, the EXP-015 refutation, and the EXP-022 quadratic refutation. The proof toolchain is pinned as follows:

Component	Identity
CaDiCaL	reported 1.7.3; Ubuntu package 1.7.4-1
CaDiCaL binary SHA-256	`7b73df0a6d9cf3c751a1948300e5baff8e82c4d39bcd88f0c063b5f5cfb8b33e`
DRAT-trim commit	`2e3b2dc0ecf938addbd779d42877b6ed69d9a985`
DRAT-trim binary SHA-256	`92f0aa9575ed519d66a99b8b1b3dde6ece4618ae4c202a3a4b200265dda0aa7a`
Z3	4.16.0; exact integer Presburger queries for EXP-023
EXP-023 query aggregate	`832c8421fe66359b8c246e3465e27de6ea7829215f892ab815e72b1f44787194`
EXP-024 campaign/audit	exact integer coefficient and dimension routes; 297 rows plus independent rebuild
EXP-027	relative-chain profiles, six exact Presburger queries, 297 formula rows, and independent reconstruction

The trusted base comprises the cited tree theorem, the finite reduction proved here, the small standard-library model checker, and DRAT-trim's verification of each proof against its exact CNF. The SAT solver need not be trusted for UNSAT. For SAT, the independently extracted model is checked without consulting solver internals.

The EXP-023 trust boundary is different. Z3's UNSAT leaves do not carry a separately checked proof object. The exact Presburger encoding, the explicit interval reductions in the proof, the separate total-graph implementation, and the finite stress campaign reduce implementation risk, but do not remove reliance on solver soundness. The one-cubic theorem is therefore labeled [**MV**], not presented as a proof-assistant theorem or a proof-carrying UNSAT result.

EXP-024 introduces no additional solver dependency. Its scripts verify arithmetic identities and the frozen hashes of the EXP-021/023 premises; Theorem 15.6 is a deductive consequence of those premises and standard homological arguments. Any defect in an imported premise would still propagate, and the theorem does not erase EXP-023's disclosed boundary.

EXP-027 uses Z3 only to adversarially negate six interval statements. The proof of Theorem 15.7 does not infer the theorem from those finite or solver results: the relative-chain identification, integral matching, explicit quadratic paths, parity obstruction, and minimal mapping cone are given above. The independent audit does not import the canonical rank implementation. The remaining trust boundary is the correctness of the frozen EXP-021/023/025 premises and the ordinary algebraic arguments displayed in the proof.

17 Scope, attribution, and open questions

Son Pham retains discovery priority for the counterexample. Professor Craig Huneke's verification is independent external evidence. The contributions proved here are Frobenius minimality, uniqueness of the normalized pair at that minimum, the separate infinite family in Theorem 9.2, and its uniform endomorphism anatomy in Theorem 10.1, pseudo-Frobenius anatomy in Theorem 11.1, exact trace/conductor formula in Theorem 12.1, stability defect in Theorem 13.1, and conductor reduction/Hilbert data in Theorem 14.1, tangent-cone depth in Theorem 15.1, and complete graded-torsion/Buchsbaum anatomy in Theorem 15.2, and the complete Noether-normalization module in Theorem 15.3, and the canonical fiber cone and type in Theorem 15.4, and its complete defining ideal in Theorem 15.5, and its exact homological edges in Theorem 15.6, and its first interior Betti strand in Theorem 15.7, together with their reproducible records.

Theorems 6.1, 7.2, 9.2, 10.1, 11.1, 12.1, 13.1, 14.1, 15.1, 15.2, 15.3, 15.4, 15.5, 15.6, and 15.7 do not classify all higher Frobenius values, prove a minimum multiplicity or embedding dimension across all counterexamples, or establish corresponding statements for arbitrary Gorenstein domains and modules. The following questions remain open:

1. How does the number of normalized rigid pairs grow above $F = 181$?
2. Is there a global minimum under multiplicity, embedding dimension, or number of minimal generators?
3. Which other Boolean-scaffold models lie on affine or residue-class families, and is the family in Theorem 9.2 maximal inside a natural Kunz face?
4. Which additional hypotheses on the endomorphism ring recover a positive theorem while excluding both the public example and the full family proved here?
5. Can the finite certificates be imported into a smaller formally verified checker?
6. Can the relative-complex method determine the remaining interior of the presentation-ring Betti table, beyond the degree-four strand in Theorem 15.7?

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